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ScienceDirect
Tetrahedron 63 (2007) 7612–7616
Tetrahedron

Studies on electrophilic reaction of Br_2 with 1,2-allenylic sulfones. A highly regio- and stereoselective synthesis of 1-phenylsulfonyl-2-bromo-1(*E*)-alken-3-ols and 1-phenylsulfonyl-2-bromo-1(*E*),3(*E*)-butadienes

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Received 6 March 2007; revised 29 April 2007; accepted 10 May 2007
Available online 13 May 2007

Abstract—The reaction of 1,2-allenyl sulfones with Br_2 afforded *E*-bromohydroxylation- or *E*-bromination–elimination products highly regio- and stereoselectively depending on the substitution pattern of the allene functionality. The five-membered intermediate *cis*-**3h** was isolated and characterized by X-ray diffraction study. The study on its reactivity of this intermediate revealed the origin of the regio- and stereoselectivity of this reaction.
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1. Introduction

Electrophilic additions of allenes[1] would be very powerful reactions in organic synthesis since two functionalities are introduced with one of the carbon–carbon double bond intact.[2] However, in order to make these methodologies synthetically attractive, the issues of regio- and stereoselectivity must be addressed.[3–6] Recently, we have demonstrated the electrophilic halocyclization reaction of functionalized allenes.[7–9] In addition, we also established the highly regio- and stereoselective halohydroxylation of 1,2-allenyl sulfoxides[10,11] and sulfides or selenides affording *E*-2-halo-3-hydroxyalkenyl sulfoxides or *Z*-2-halo-3-hydroxyalkenyl sulfides or selenides, respectively.[12–14] Although rationales for the stereoselectivity observed in these reactions were provided, they all lack direct experimental proof. In addition, under the same reaction conditions used for 1,2-allenyl sulfoxides,[10,11] no reaction occurred with 1,2-allenyl sulfones probably due to the relatively stronger electron-withdrawing ability of the sulfone functionality.[15] In this paper, we wish to report the realization of highly regio- and stereoselective bromohydroxylation reaction of the electron-deficient 1,2-allenyl sulfones with Br_2 leading to *E*-2-bromo-3-hydroxyalkenyl sulfones or *E*-2-bromo-1,3-alkadienyl sulfones depending on the substitution pattern of the allene moiety. The intermediate responsible for the regio- and stereoselectivity observed was isolated and characterized by the X-ray diffraction study.

2. Results and discussion

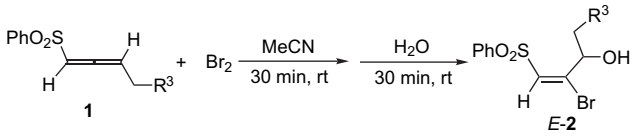
After some screening, it was observed that the reaction of 3-monoalkyl substituted 1,2-allenyl phenyl sulfones **1** with Br_2 in MeCN followed by the addition of H_2O afforded *E*-2-bromo-3-hydroxyalkenyl sulfones **2** with very high regio- and stereoselectivity, which were established by the X-ray diffraction study of **2c**.[16] Some of the most typical results of this bromohydroxylation reaction are summarized in Table 1 (Fig. 1).

In order to search the origin of the regio- and stereoselectivity, after some trial and error, 1,2-butadienyl *p*-bromophenyl sulfone **1h** was prepared and its reaction with 1.2 equiv of Br_2 in MeCN at 0 °C afforded the expected product **2h** in 85% yield. However, when 2.0 equiv of Br_2 were applied, after careful handling, a five-membered intermediate with Br_3^- as the counter ion, i.e., *cis*-**3h** was isolated with the structure

Keywords: 1,2-Allenylic sulfones; Bromine; Bromohydroxylation; Mechanistic study.
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0040–4020/$ - see front matter © 2007 Published by Elsevier Ltd.
doi:10.1016/j.tet.2007.05.039

Table 1. *E*-Bromohydroxylation reaction of 1,2-allenyl sulfones



Entry	R^3	Br_2 (equiv)	Yield of *E*-**2** (%)
1[a]	H (**1a**)	1.2	85 (**2a**)
2	CH_3 (**1b**)	1.2	68 (**2b**)
3	C_2H_5 (**1c**)	2.0	60 (**2c**)
4	*n*-C_3H_7 (**1d**)	2.0	63 (**2d**)
5	*n*-C_4H_9 (**1e**)	1.2	63 (**2e**)
6	*n*-C_5H_{11} (**1f**)	1.5	58 (**2f**)
7[b]	*n*-C_5H_{11} (**1f**)	1.5	53 (**2f**)
8	Ph (**1g**)	2.0	71 (**2g**)

[a] The reaction was conducted at 0 °C.
[b] Solvent: MeCN/EtOH=40:1, 1.0 equiv of H_2O.

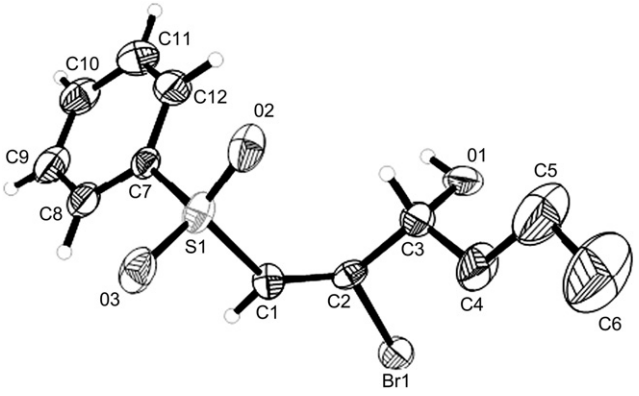


Figure 1. ORTEP representation of **2c**.

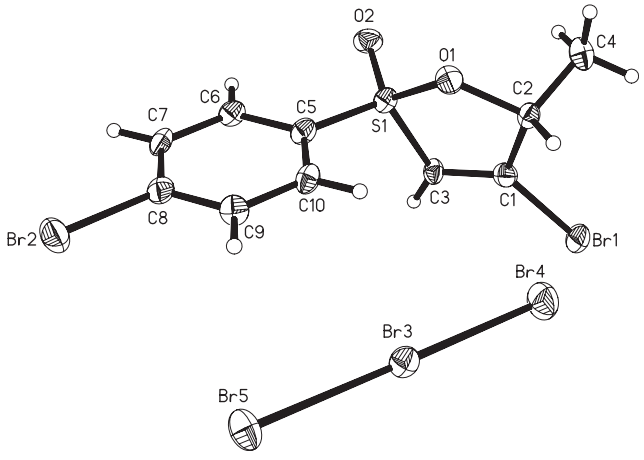
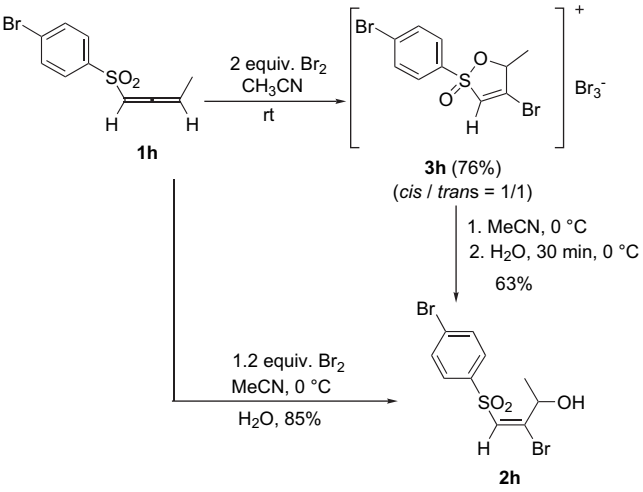


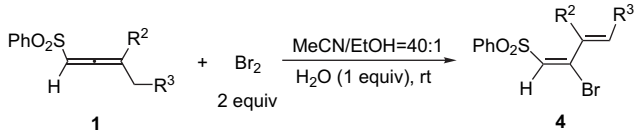
Figure 2. ORTEP representation of *cis*-**3h**.

being established by X-ray diffraction study[17] (Fig. 2). Its reaction with H_2O in MeCN afforded **2h** as the single product cleanly (Scheme 1)! The intermediacy of **3h** explains the regio- and stereoselectivity and provides strong proof for the rationale that the sulfoxide functionality was involved to form a similar five-membered intermediate in the highly regio- and stereoselective *E*-halohydroxylation of 1,2-allenyl sulfoxides (Fig. 2).[11,18]



Scheme 1.

Table 2. Synthesis of 1-phenylsulfonyl-2-bromo-1,3-butadienes (*E*,*E*)-4 via the reactions of 1,2-allenyl sulfones



Entry	R^2	R^3	Yield of *E*,*E*-**4**	*E*/*Z*	
				C=C (1)	C=C (3)
1	CH_3	H (**1i**)	76 (**4i**)	>99:1	—
2[a]	$(CH_2)_4$ (**1j**)		63 (**4j**)	>99:1	>99:1
3	Ph	CH_3 (**1k**)	80 (**4k**)	>99:1	>99:1

[a] The reaction was conducted at −40 °C.

Furthermore, when 3,3-disubstituted 1,2-allenyl sulfones were applied, the reaction under the conditions given in Table 1 was either not clean or low-yielding. After screening, it was observed that the reaction in MeCN/EtOH (40:1) *in the presence of 1 equiv of water* afforded 2-bromo-1,3-alkadienyl sulfones (*E*,*E*)-**4** highly stereoselectively, indicating a further elimination reaction following the bromination yielding the second carbon–carbon double bond highly stereoselectively[12–14] (Table 2). However, under this set of conditions, the reaction of **1f** still afforded the bromohydroxylation product *E*-**2f** (entry 7, Table 1).

3. Conclusion

In conclusion, a highly regio- and stereoselective *E*-bromohydroxylation or *E*-bromination–elimination reaction has been established. The reaction pathway depends largely on the substitution pattern of the allene functionality. The intermediacy of these reactions has been established by the isolation, X-ray diffraction characterization, and the reactivity study of the five-membered intermediate *cis*-**3h**, which provide a proof for the rationale that the participation of sulfoxide or sulfone functionality[18] forming a five-membered intermediate determines the stereoselectivity of these halohydroxylation reactions. Further studies in this area are being conducted in our laboratory.

4. Experimental section

4.1. Synthesis of *E*-2-bromo-3-hydroxyalkenyl sulfones via the *E*-bromo-hydroxylation reaction of 1,2-allenyl sulfones

4.1.1. 3-Bromo-4-(phenylsulfonyl)but-3(*E*)-en-2-ol (2a). Typical procedure: to a solution of **1a** (58.5 mg, 0.3 mmol) in 3.3 mL of MeCN was added with stirring 0.72 mL of Br_2 (0.5 M in MeCN, 0.36 mmol) at 0 °C. After the addition, the reaction was stirred at 0 °C for 0.5 h. The resulting mixture was treated with 6 mL of water and stirred for another 0.5 h. Then it was extracted with diethyl ether (50+25×2 mL), washed with brine, and dried over *anhydrous* Na_2SO_4. Evaporation and column chromatography on silica gel (petroleum ether/ethyl acetate=5:1) afforded **2a** (73.9 mg, 85%) as a solid, mp 78–79 °C (hexane and diethyl ether). ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.91 (d, J=8.4 Hz, 2H), 7.67 (t, J=8.0 Hz, 1H), 7.58 (t, J=8.0 Hz, 2H), 6.78 (s, 1H), 5.51 (t, J=6.6 Hz, 1H), 2.78 (d, J=6.8 Hz, 1H), 1.37 (d, J=6.0 Hz, 3H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 150.6, 140.1, 134.1, 131.3, 129.6, 127.4, 65.4, 22.2; IR (neat) ν (cm^{-1}) 3479, 3046, 1601, 1302, 1120; MS (70 eV, EI) m/z (%): 293 (M^+(^{81}Br), 9.91), 291 (M^+(^{79}Br), 9.92), 77 (100). Elemental analysis Calcd for $C_{10}H_{11}BrO_3S$: C, 41.25; H, 3.81. Found: C, 41.21; H, 3.78.

4.1.2. 2-Bromo-1-(phenylsulfonyl)pent-1(*E*)-en-3-ol (2b). The reaction of 62.4 mg (0.3 mmol) of **1b** and 0.72 mL (0.5 M in MeCN, 0.36 mmol) of Br_2 at rt afforded 62.0 mg (68%) of **2b** as an oil. ^{1}H NMR δ 7.91 (d, J=6.8 Hz, 2H), 7.66 (t, J=6.8 Hz, 1H), 7.57 (t, J=6.8 Hz, 2H), 6.83 (s, 1H), 5.33 (t, J=6.8 Hz, 1H), 2.74 (br s, 1H), 1.70–1.79 (m, 1H), 1.56–1.65 (m, 1H), 0.93 (t, J=7.4 Hz, 3H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 149.5, 140.3, 134.1, 132.5, 129.6, 127.5, 70.2, 29.1, 9.4; IR (neat) ν (cm^{-1}) 3487, 3043, 1601, 1447, 1310, 1115; MS (70 eV, EI) m/z (%): 277 (M^+(^{81}Br)−C_2H_5, 5.23), 275 (M^+(^{79}Br)−C_2H_5, 8.71), 151 (100). Elemental analysis Calcd for $C_{11}H_{13}BrO_3S$: C, 43.29; H, 4.27. Found: C, 43.29; H, 4.36.

4.1.3. 2-Bromo-1-(phenylsulfonyl)hex-1(*E*)-en-3-ol (2c). The reaction of 66.3 mg (0.3 mmol) of **1c** and 1.2 mL (0.5 M in MeCN, 0.6 mmol) of Br_2 at rt afforded 57.2 mg (60%) of **2c** as an oil. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.91 (d, J=7.8 Hz, 2H), 7.66 (t, J=7.8 Hz, 1H), 7.57 (t, J=7.8 Hz, 2H), 6.80 (s, 1H), 5.40 (t, J=6.2 Hz, 1H), 2.74 (br s, 1H), 1.67–1.75 (m, 1H), 1.40–1.66 (m, 2H), 1.23–1.35 (m, 1H), 0.93 (t, J=7.0 Hz, 3H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 149.9, 140.2, 134.0, 131.9, 129.5, 127.4, 68.7, 37.9, 18.2, 13.7; IR (KBr) ν (cm^{-1}) 3492, 3044, 1604, 1309, 1152; MS (70 eV, EI) m/z (%): 303 (M^+(^{81}Br)−OH, 0.73), 301 (M^+(^{79}Br)−OH, 0.83), 77 (100); HRMS calcd for $C_{12}H_{15}{}^{79}BrO_3S$ (M^+): 317.9925, found: 317.9938.

4.1.4. 2-Bromo-1-(phenylsulfonyl)hept-1(*E*)-en-3-ol (2d). The reaction of 71.5 mg (0.3 mmol) of **1d** and 1.2 mL (0.5 M in MeCN, 0.6 mmol) of Br_2 at rt afforded 62.8 mg (63%) of **2d** as an oil. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.90 (d, J=7.8 Hz, 2H), 7.66 (t, J=7.8 Hz, 1H), 7.55 (t, J=7.8 Hz, 2H), 6.81 (s, 1H), 5.38 (t, J=6.2 Hz, 1H), 2.96 (br s, 1H), 1.63–1.74 (m, 1H), 1.45–1.59 (m, 1H), 1.18–1.42 (m, 4H), 0.88 (t, J=7.0 Hz, 3H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 150.0, 140.2, 134.0, 132.0, 129.5, 127.4, 68.7, 35.5, 26.9, 22.3, 13.8; IR (KBr) ν (cm^{-1}) 3490, 3044, 1603, 1447, 1310, 1152; MS (70 eV, EI) m/z (%): 335 (M^++1(^{81}Br), 0.58), 333 (M^++1(^{79}Br), 0.80), 125 (100); HRMS calcd for $C_{13}H_{17}{}^{79}BrO_3S$ (M^+): 332.0082, found: 332.0099.

4.1.5. 2-Bromo-1-(phenylsulfonyl)oct-1(*E*)-en-3-ol (2e). The reaction of 73.8 mg (0.3 mmol) of **1e** and 0.72 mL (0.5 M in MeCN, 0.36 mmol) of Br_2 at rt afforded 64.0 mg (63%) of **2e** as an oil. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.90 (d, J=7.2 Hz, 2H), 7.65 (t, J=7.2 Hz, 1H), 7.56 (t, J=7.2 Hz, 2H), 6.81 (s, 1H), 5.38 (t, J=6.8 Hz, 1H), 2.84 (br s, 1H), 1.65–1.72 (m, 1H), 1.49–1.59 (m, 1H), 1.35–1.45 (m, 1H), 1.20–1.34 (m, 5H), 0.86 (t, J=6.2 Hz, 3H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 150.0, 140.3, 134.0, 132.0, 129.5, 127.4, 68.9, 35.8, 31.3, 24.5, 22.3, 13.9; IR (KBr) ν (cm^{-1}) 3492, 3039, 1603, 1309, 1152; MS (70 eV, EI) m/z (%): 277 (M^+(^{81}Br)−C_5H_{11}, 25.5), 275 (M^+(^{79}Br)−C_5H_{11}, 24.9), 125 (100); HRMS calcd for $C_{14}H_{19}BrO_3SNa$ (M^++Na): 371.0115 (^{81}Br), 369.0136 (^{79}Br), found: 371.0094 (^{81}Br), 369.0116 (^{79}Br).

4.1.6. 2-Bromo-1-(phenylsulfonyl)non-1(*E*)-en-3-ol (2f). The reaction of 79.3 mg (0.3 mmol) of **1f** and 1.2 mL (0.5 M in MeCN, 0.6 mmol) of Br_2 at rt afforded 62.6 mg (58%) of **2f** as an oil. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.91 (d, J=7.8 Hz, 2H), 7.65 (t, J=7.8 Hz, 1H), 7.56 (t, J=7.8 Hz, 2H), 6.81 (s, 1H), 5.38 (t, J=6.8 Hz, 1H), 2.79 (br s, 1H), 1.64–1.75 (m, 1H), 1.49–1.60 (m, 1H), 1.25–1.44 (m, 8H), 0.87 (t, J=6.4 Hz, 3H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 149.9, 140.3, 134.0, 132.1, 129.5, 127.4, 69.0, 35.9, 31.5, 28.9, 24.8, 22.4, 14.0; IR (KBr) ν (cm^{-1}) 3491, 2928, 1604, 1310, 1152; MS (70 eV, EI) m/z (%): 363 (M^++1(^{81}Br), 0.52), 361 (M^++1(^{79}Br), 0.56), 125 (100); HRMS calcd for $C_{15}H_{21}{}^{79}BrO_3S$ (M^+): 360.0395, found: 360.0407.

4.1.7. 2-Bromo-1-(phenylsulfonyl)non-1(*E*)-en-3-ol (2f). To a solution of **1f** (80.0 mg, 0.3 mmol) in 2.8 mL of MeCN were added with stirring 0.1 mL of EtOH and 5.4 μL (5.4 mg, 0.3 mmol) of H_2O. Then 1.2 mL (0.5 M in MeCN, 0.6 mmol) of Br_2 was added at rt. After the addition, the reaction was stirred at rt for 0.5 h. The resulting mixture was treated with 6 mL of water and stirred for another 0.5 h. Then it was extracted with diethyl ether (50+25×2 mL), washed with brine, and dried over *anhydrous* Na_2SO_4. Evaporation and column chromatography on silica gel (petroleum ether/ethyl acetate=5:1) afforded *E*-**2f** (58.5 mg, 53%).

4.1.8. 3-Bromo-1-phenyl-4-(phenylsulfonyl)but-3(*E*)-en-2-ol (2g). The reaction of 80.8 mg (0.3 mmol) of **1g** and 1.2 mL (0.5 M in MeCN, 0.6 mmol) of Br_2 at rt afforded 77.5 mg (71%) of **2g** as an oil. ^{1}H NMR (400 MHz, $CDCl_3$) δ 7.71 (d, J=7.6 Hz, 2H), 7.63 (t, J=7.6 Hz, 1H), 7.51 (t, J=7.6 Hz, 2H), 7.25–7.34 (m, 5H), 6.77 (s, 1H), 5.76 (t, J=6.6 Hz, 1H), 3.01 (d, J=7.2 Hz, 2H), 2.72 (s, 1H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 148.3, 139.9, 135.8, 133.9, 132.3, 129.7, 129.5, 128.6, 127.4, 127.0, 69.7, 42.4; IR (KBr) ν (cm^{-1}) 3489, 3031, 1603, 1308, 1152; MS (70 eV, EI) m/z (%): 277 (M^+(^{81}Br)−Bn, 10.5), 275 (M^+(^{79}Br)−Bn, 10.5), 91 (100); HRMS calcd for

$C_{16}H_{15}BrNaO_3S$ (M^++Na): 390.9802 (^{81}Br), 388.9823 (^{79}Br), found: 390.9793 (^{81}Br), 388.9812 (^{79}Br).

4.1.9. 3-Bromo-4-(4′-bromophenylsulfonyl)but-3(*E*)-en-2-ol (2h). The reaction of 54.1 mg (0.2 mmol) of **1h** and 0.48 mL (0.5 M in MeCN, 0.24 mmol) of Br_2 at 0 °C afforded 53.4 mg (73%) of **2h** as a solid, mp 98–100 °C (hexane and diethyl ether). ^1H NMR (400 MHz, $CDCl_3$) δ 7.78 (d, J=9.2 Hz, 2H), 7.72 (d, J=9.2 Hz, 2H), 6.75 (s, 1H), 5.54 (q, J=6.3 Hz, 1H), 2.72 (br s, 1H), 1.37 (d, J=6.4 Hz, 3H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 151.3, 139.2, 132.9, 130.9, 129.5, 129.0, 65.5, 22.3; IR (KBr) ν (cm^{-1}) 3484, 2972, 1598, 1572, 1314, 1148; MS (70 eV, EI) *m/z* (%): 357 (M^+(^{81}Br, ^{81}Br)−CH_3, 2.6), 355 (M^+(^{81}Br, ^{79}Br)−CH_3, 3.9), 353 (M^+ (^{79}Br, ^{79}Br)−CH_3, 2.3), 149 (100). Elemental analysis Calcd for $C_{10}H_{10}Br_2O_3S$: C, 32.46; H, 2.72. Found: C, 32.60; H, 2.81.

4.2. Synthesis of 1-phenylsulfonyl-2-bromo-1,3-butadienes via the reactions of 3,3-disubstituted 1,2-allenyl sulfones

4.2.1. *E*-(2-Bromo-3-methylbuta-1,3-dienyl) phenyl sulfone (4i). Typical procedure: to a solution of **1i** (62.1 mg, 0.3 mmol) in 2.8 mL of MeCN were added with stirring 0.1 mL of EtOH and 5.4 μL (5.4 mg, 0.3 mmol) of H_2O. Then 1.2 mL (0.5 M in MeCN, 0.6 mmol) of Br_2 was added at rt. After being stirred for 5 min, the resulting mixture was treated with 6 mL of water, extracted with diethyl ether (30+25×2 mL), washed with brine, and dried over *anhydrous* Na_2SO_4. Evaporation and column chromatography on silica gel (petroleum ether/ethyl acetate=20:1) afforded **4i** (65.3 mg, 76%) as an oil. ^1H NMR (400 MHz, $CDCl_3$) δ 7.87 (d, J=7.8 Hz, 2H), 7.63 (t, J=7.2 Hz, 1H), 7.53 (t, J=7.6 Hz, 2H), 6.83 (s, 1H), 5.15 (s, 2 H), 1.82 (s, 3H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 141.9, 140.7, 140.6, 133.7, 132.7, 129.2, 128.0, 119.6, 20.6; IR (neat) ν (cm^{-1}) 3051, 1600, 1325, 1152; MS (70 eV, EI) *m/z* (%): 285 (M^+(^{79}Br), 1.2), 287 (M^+(^{81}Br), 1.0), 161 (100); HRMS calcd for $C_{11}H_{11}BrNaO_2S$ (M^++Na): 310.9540 (^{81}Br), 308.9561 (^{79}Br), found: 310.9549 (^{81}Br), 308.9569 (^{79}Br).

4.2.2. *E*-1-Phenylsulfonyl-2-bromo-2-cyclohexenylethene (4j). The reaction of 49.1 mg (0.2 mmol) of **1j**, 67.5 μL of EtOH, 3.6 μL of H_2O, and 0.8 mL (0.5 M in MeCN, 0.4 mmol) of Br_2 at −40 °C afforded 42.1 mg (65%) of **4j** as an oil. ^1H NMR (400 MHz, $CDCl_3$) δ 7.83 (d, J=7.6 Hz, 2H), 7.62 (t, J=7.6 Hz, 1H), 7.53 (t, J=7.6 Hz, 2H), 6.87 (s, 1H), 5.94–5.96 (m, 1H), 2.03–2.08 (m, 2H), 1.86–1.88 (m, 2H), 1.45–1.50 (m, 4H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 143.5, 141.1, 134.3, 133.5, 133.2, 132.5, 129.0, 127.9, 26.3, 25.2, 21.5, 20.9; IR (neat) ν (cm^{-1}) 3049, 1641, 1596, 1320, 1149; MS (70 eV, EI) *m/z* (%): 328 (M^+(^{81}Br), 5.8), 326 (M^+(^{79}Br), 5.6), 105 (100); HRMS calcd for $C_{14}H_{15}BrNaO_2S$ (M^++Na): 350.9853 (^{81}Br), 348.9874 (^{79}Br), found: 350.9840 (^{81}Br), 348.9859 (^{79}Br).

4.2.3. 1-Phenylsulfonyl-2-bromo-3-phenyl-1(*E*),3(*E*)-pentadiene (4k). The reaction of **1k** (56.2 mg, 0.2 mmol) in 1.9 mL of MeCN, 67.5 μL of EtOH, and 3.6 μL of H_2O (3.6 mg, 0.2 mmol) with 0.8 mL of Br_2 (0.5 M in MeCN, 0.4 mmol) at rt afforded **4k** (57.2 mg, 80%). The reaction mixture was submitted to chromatography on silica gel without workup. Compound **4k**: oil; ^1H NMR (400 MHz, $CDCl_3$) δ 7.67 (d, J=8.0 Hz, 2H), 7.53 (t, J=8.0 Hz, 1H), 7.40 (t, J=8.0 Hz, 2H), 7.26–7.31 (m, 5H), 7.14 (s, 1H), 6.28 (q, J=7.0 Hz, 1H), 1.84 (d, J=7.0 Hz, 3H); ^{13}C NMR (100 MHz, $CDCl_3$) δ 139.8, 137.2, 136.8, 135.8, 134.5, 133.7, 129.1, 128.6, 128.5, 127.9, 127.7, 125.8, 16.0; IR (neat) ν (cm^{-1}) 3045, 1631, 1592, 1327, 1153; MS (70 eV, EI) *m/z* (%): 364 (M^+(^{81}Br), 0.72), 362 (M^+(^{79}Br), 1.0), 103 (100); HRMS calcd for $C_{17}H_{15}BrNaO_2S$ (M^++Na): 386.9852 (^{81}Br), 384.9874 (^{79}Br), found: 386.9852 (^{81}Br), 384.9872 (^{79}Br).

4.3. Isolation and reaction of the intermediate *cis*-3h and study of the mechanism

4.3.1. Isolation and reaction of intermediate *cis*-3h. To a solution of **1h** (546 mg, 2.0 mmol) in 4.0 mL of anhydrous MeCN was added with stirring 8.0 mL (0.5 M in anhydrous MeCN, 4.0 mmol) of Br_2 at rt. After evaporation of the solvent, an orange solid was collected by filtration. It was washed with 10×3 mL of anhydrous diethyl ether and dried under vacuum to afford 940.7 mg (79%, *cis*/*trans* isomer 1:1) of **3h**. This solid was dissolved in anhydrous MeCN at 40 °C and then recrystallized at 0 °C to afford pure *cis*-**3h** as a solid, mp 103–105 °C (MeCN) (decomposition). ^1H NMR (400 MHz, CD_3CN) δ 8.07 (d, J=8.8 Hz, 2H), 8.01 (d, J=8.8 Hz, 2H), 7.68 (d, J=1.6 Hz, 1H), 6.45 (dq, J=7.2 and 1.6 Hz, 1H), 1.99 (d, J=7.2 Hz, 3H); IR (KBr) ν (cm^{-1}) 3083, 1565, 1308, 1278; MS (70 eV, EI) *m/z* (%): 355 (M^+(^{81}Br, ^{81}Br), 4.8), 353 (M^+(^{79}Br, ^{81}Br), 2.8), 351 (M^+(^{79}Br, ^{79}Br), 0.27), 149 (100); HRMS calcd for $C_{10}H_9BrO_3S$ (M^+): 354.8649 (^{81}Br, ^{81}Br), 352.8669 (^{79}Br, ^{81}Br), 350.8690 (^{79}Br, ^{79}Br), found: 354.8645 (M^+(^{81}Br, ^{81}Br), 352.8668 (M^+(^{79}Br, ^{81}Br), 350.8689 (^{79}Br, ^{79}Br).

4.3.2. The reaction of 3h with H_2O affording 2h. To a solution of **3h** (118.6 mg, 0.2 mmol, *cis*/*trans* ratio: 1:1) in 2.7 mL of MeCN was added with stirring 4.0 mL of water at 0 °C. After being stirred for 0.5 h at 0 °C as monitored by TLC, it was extracted with diethyl ether (30+25×2 mL), washed with brine, and dried over *anhydrous* Na_2SO_4. Evaporation and column chromatography on silica gel (petroleum ether/ethyl acetate=4:1) afforded **2h** (46.3 mg, 63%).

Acknowledgements

We gratefully acknowledge the National Natural Science Foundation of China (No. 20572093), Zhejiang Provincial Natural Science Foundation of China (Y 404262), and Cheung Kong Scholar Program for financial support. Shengming Ma is jointly appointed by Zhejiang University and Shanghai Institute of Organic Chemistry. This work was conducted at Zhejiang University. We thank Shau Lu in this group for reproducing the results presented entries 2 and 6 in Table 1 and entry 1 in Table 2.

Supplementary data

Supplementary data associated with this article can be found in the online version, at doi:10.1016/j.tet.2007.05.039.

References and notes

1. For most recent reviews on the chemistry of allenes, see: (a) Krause, N.; Hashmi, A. S. K. *Modern Allene Chemistry*; Wiley-VCH: Weinheim, 2004; (b) Ma, S. *Chem. Rev.* **2005**, *105*, 2829.
2. For a review on electrophilic addition of allenes, see: Ma, S. Chapter 10 in Ref. 1a.
3. Katrizky, A. R.; Feng, D. *J. Org. Chem.* **1997**, *62*, 715.
4. Kocienski, P.; Whitby, R. *Synthesis* **1991**, 1029.
5. Hiof, H.; Natsuas, K. *Liebigs Ann. Chem.* **1988**, 705.
6. Mueller, W. H.; Butler, P. E. *J. Org. Chem.* **1968**, *33*, 1533.
7. Ma, S.; Wu, S. *J. Org. Chem.* **1999**, *64*, 9314.
8. Ma, S.; Wu, S. *Tetrahedron Lett.* **2001**, *42*, 4075.
9. Fu, C.; Ma, S. *Eur. J. Org. Chem.* **2005**, *18*, 3942.
10. Ma, S.; Wei, Q.; Wang, H. *Org. Lett.* **2000**, *2*, 3893.
11. Ma, S.; Ren, H.; Wei, Q. *J. Am. Chem. Soc.* **2003**, *125*, 4817 and the references cited therein.
12. Ma, S.; Hao, X.; Huang, X. *Org. Lett.* **2003**, *5*, 1217.
13. Ma, S.; Hao, X.; Meng, X.; Huang, X. *J. Org. Chem.* **2004**, *69*, 5720.
14. Ma, S.; Hao, X.; Huang, X. *Chem. Commun.* **2003**, 1082.
15. For the electrophilic addition of Br_2 or I_2 with 1,2-propadienyl phenyl sulfone, see: (a) Padwa, A.; Murphree, S.; Yeske, P. *J. Org. Chem.* **1990**, *55*, 4241; (b) Padwa, A.; Ishida, M. *Tetrahedron Lett.* **1991**, *32*, 5673; For the electrophilic addition of Br_2 with bisallenic sulfones forming γ-sultines, see: (c) Braverman, S.; Duar, Y. *J. Am. Chem. Soc.* **1983**, *105*, 1061; (d) Braverman, S.; Reisman, D. *J. Am. Chem. Soc.* **1977**, *99*, 605.
16. Crystal data for compound **2c**: $C_{12}H_{15}BrO_3S$, M_w 319.21, monoclinic, *C*2/*c*, Mo Kα, final *R* indices [$I>2\sigma(I)$], R=0.0617, *wR*2=0.1502, a=35.063(4) Å, b=8.3963(9) Å, c=25.747(3) Å, α=90°, β=132.425(5)°, γ=90°, V=5595.2(11), Z=16, number of reflections measured/unique 16035/6260 (R_{int}=0.063) number of observations 2896 [$I>2\sigma(I)$], parameter 342, CCDC 614239.
17. Crystal data for compound *cis*-**3h**: $C_{10}H_9Br_5O_2S$, M_w 592.78, monoclinic, *P*2(1)/*n*, Mo Kα, final *R* indices [$I>2\sigma(I)$], R=0.0825, *wR*2=0.1776, a=9.8897(11) Å, b=10.7078(12) Å, c=14.8225(16) Å, α=90°, β=93.634(2)°, γ=90°, V=1566.5(3), Z=4, number of reflections measured/unique 9049/3414 (R_{int}=0.1978) number of observations 2152 [$I>2\sigma(I)$], parameter 164, CCDC 614241.
18. (a) Cadogan, J. I. G.; Cameron, D. K.; Gosney, I.; Millar, J. R. A.; Newlands, S. F.; Reed, D. *J. Chem. Soc., Perkin Trans. 2* **1996**, 2309; (b) Cossu, S.; de Lucchi, O.; Dilillo, F. *Gazz. Chim. Ital.* **1991**, *121*, 533.